

Math 3500 Homework: Applications of Volume Integration

This homework assignment covers Section 7.4 from Shifrin. Please print out this template^a and use it to submit your work, putting each answer in the corresponding work area for the problem.

1. Use a dark (#2) pencil or a blue or black pen to do your work.
 2. To help preserve blind grading, please *don't* write your name on the homework.
 3. Some of the questions have a “final answer” area. If one is provided, please use it – it helps us group correct answers (and common mistakes) together in Gradescope.
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1. Recall that $B(\mathbf{0}, a) \subset \mathbb{R}^3$ is the ball of radius a centered at $\mathbf{0}$ in $\mathbb{R}^3$. What is the average distance between the points in $B(\mathbf{0}, a)$ and...
(1) (cf. Shifrin 7.4.2) ...the center of the ball?

^aThere are printer kiosks around campus that you can use with your student ID if you need to. See this link for details.

(2) (cf. Shifrin 7.4.4) ... a (fixed) point $\mathbf{x}_0$ on the surface of the ball?^b



^bBy symmetry, the answer will be the same for any choice of $\mathbf{x}_0$. For instance, you're welcome to let $\mathbf{x}_0 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ if you'd like to.

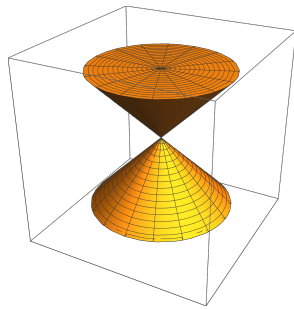
2. (cf. Shifrin 7.4.9) Suppose that Ω is the solid hemisphere defined by

$$x^2 + y^2 + z^2 \leq a^2, \quad z \geq 0.$$

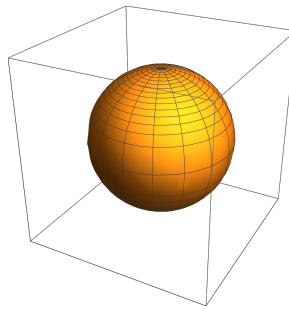
Find the center of mass of Ω (in terms of a).

3. (cf. Shifrin 7.4.13) Suppose that Ω is the ball $B(\mathbf{0}, a) \subset \mathbb{R}^3$ and the density of Ω is given by $\delta(\mathbf{x}) = \|\mathbf{x}\|$. Find the moment of inertia of Ω around the z -axis.

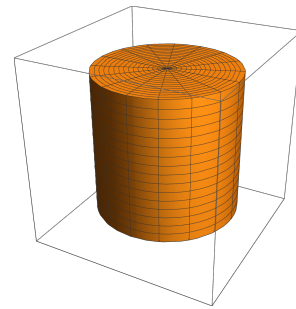
4. (1) Consider the following solids:



Cone
(height $2a$, radius a)

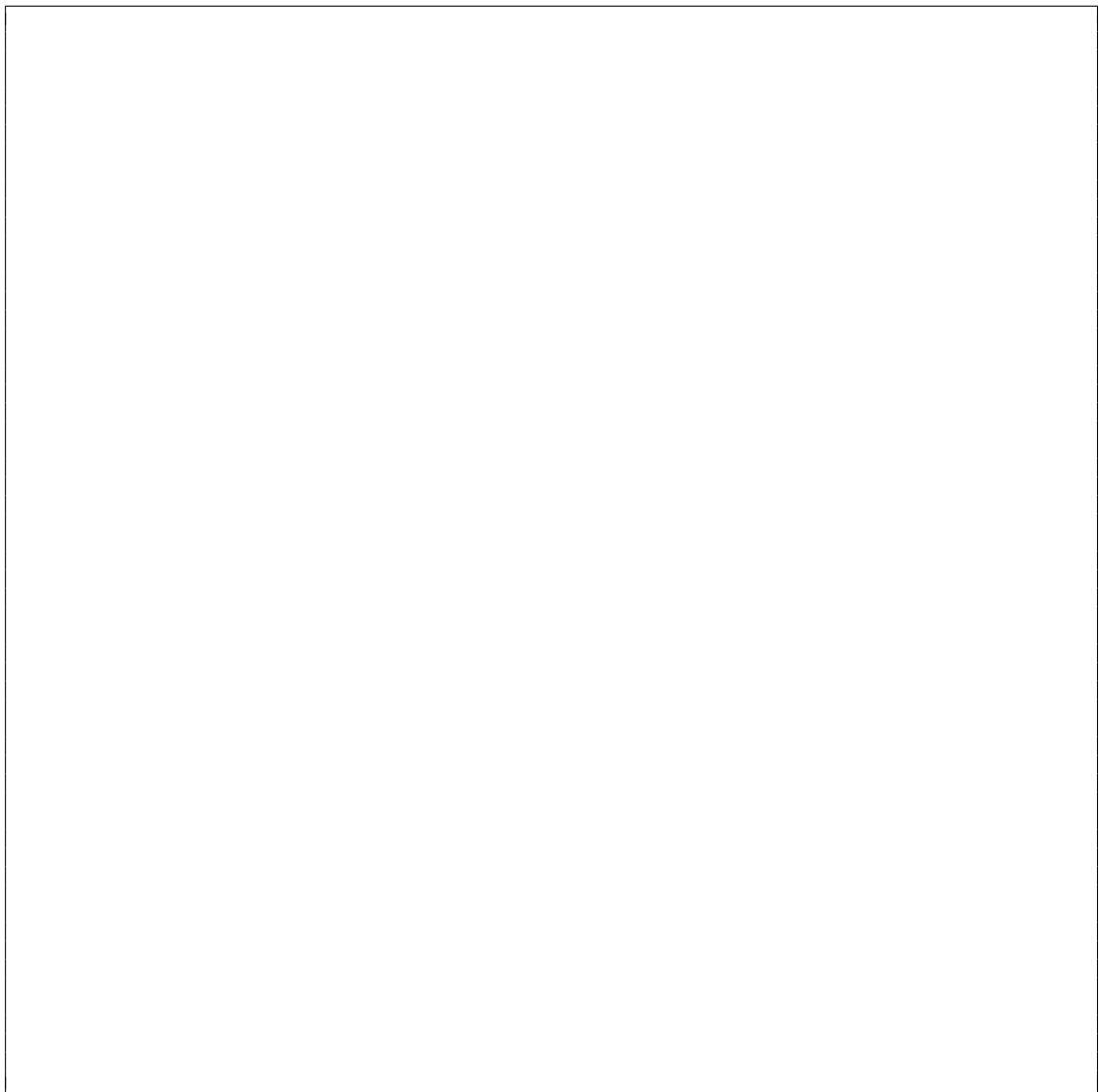


Sphere
(radius a)

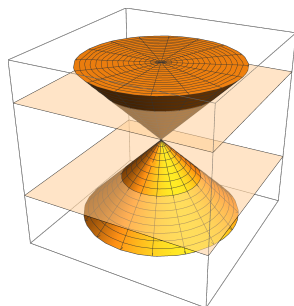


Cylinder
(height $2a$, radius a)

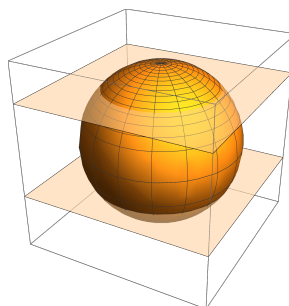
Look up the volume of each solid and prove the weak form of *Archimedes' principle*: the volume of the entire cone, the entire sphere, and the entire cylinder are in the ratio $1 : 2 : 3$.



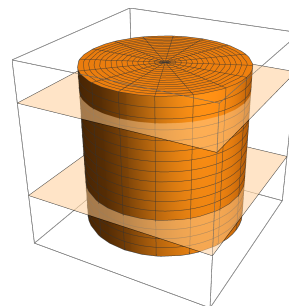
- (2) Now consider the portion of each solid between two horizontal planes at heights $-a \leq b < c \leq a$, as in the pictures below:



Cone
(height $2a$, radius a)



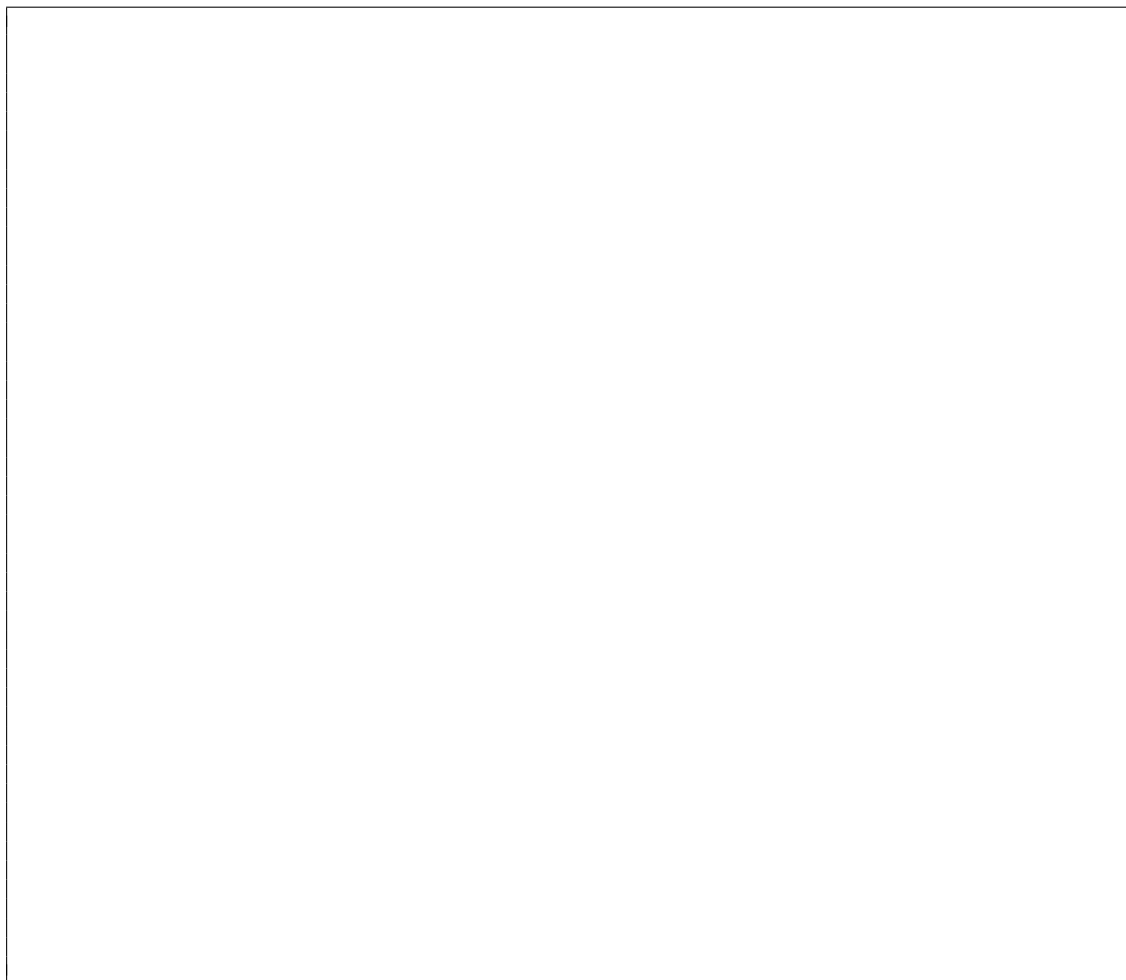
Sphere
(radius a)

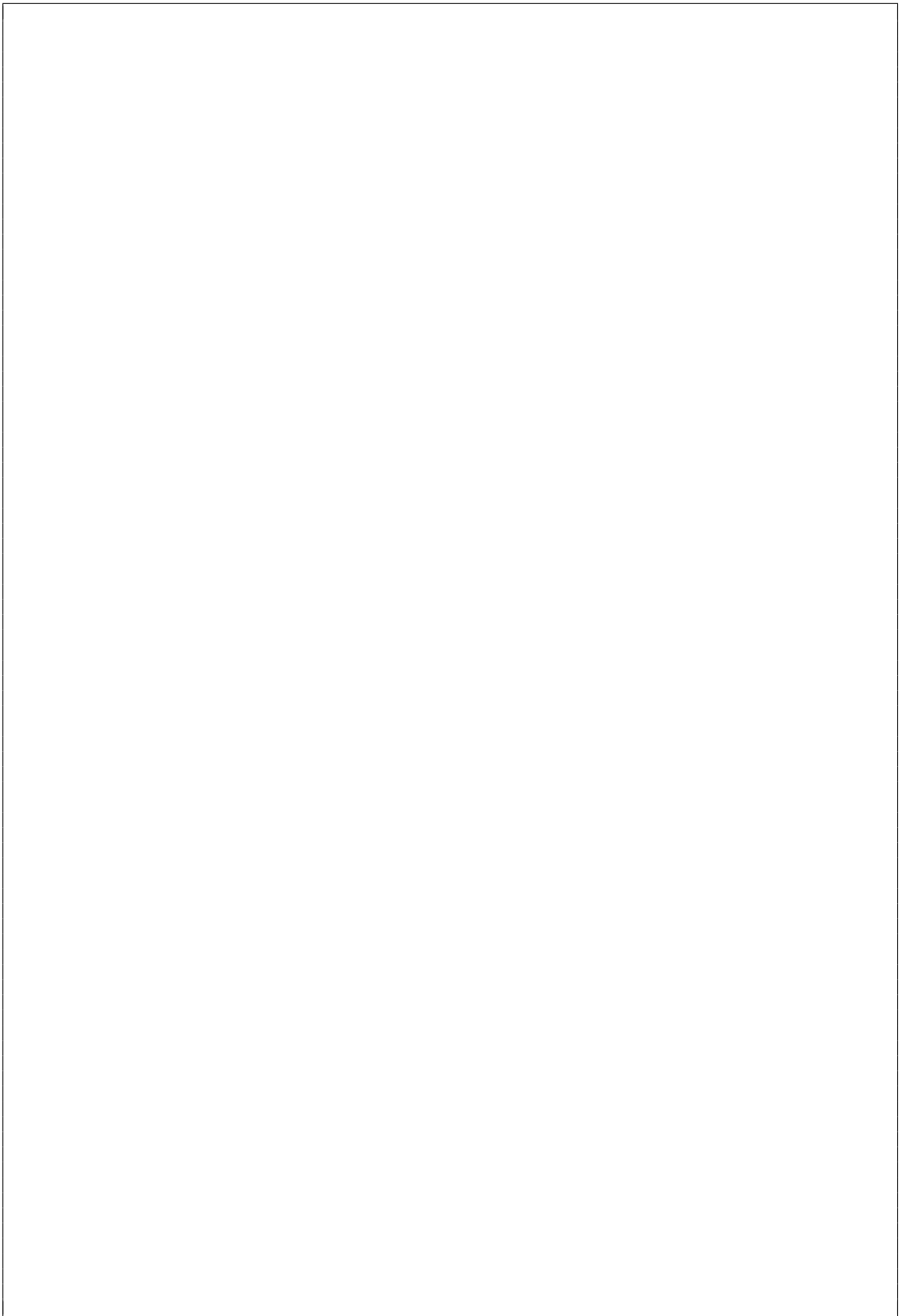


Cylinder
(height $2a$, radius a)

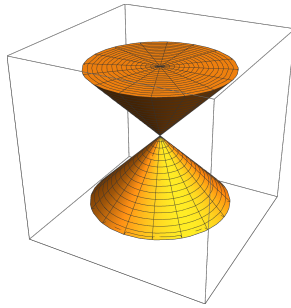
Let $\text{Cone}(b, c)$, $\text{Sphere}(b, c)$ and $\text{Cyl}(b, c)$ denote the portions of the cone, sphere, and cylinder between the planes. Use integration to prove the strong form of Archimedes' principle:

$$\text{VolCone}(b, c) + \text{VolSphere}(b, c) = \text{VolCyl}(b, c)$$

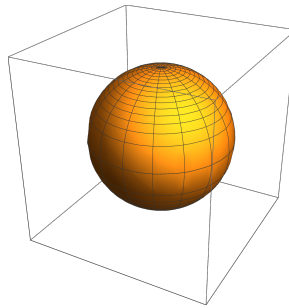




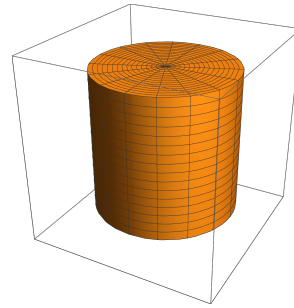
5. (cf. Shifrin 7.4.16) Consider again the following solids:



Cone
(height $2a$, radius a)



Sphere
(radius a)



Cylinder
(height $2a$, radius a)

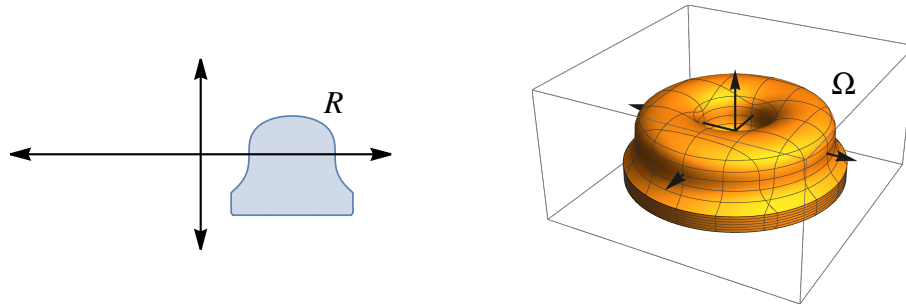
Find the moment of inertia around the z -axis of the (solid) cone, sphere, and cylinder, assuming that each object has uniform density ρ . In each case, express your answer in the form $I = kma^2$ where m is the mass of the object and k is a constant.

(1) The moment of inertia of the cone around the z -axis is:

(2) The moment of inertia of the sphere around the z -axis is:

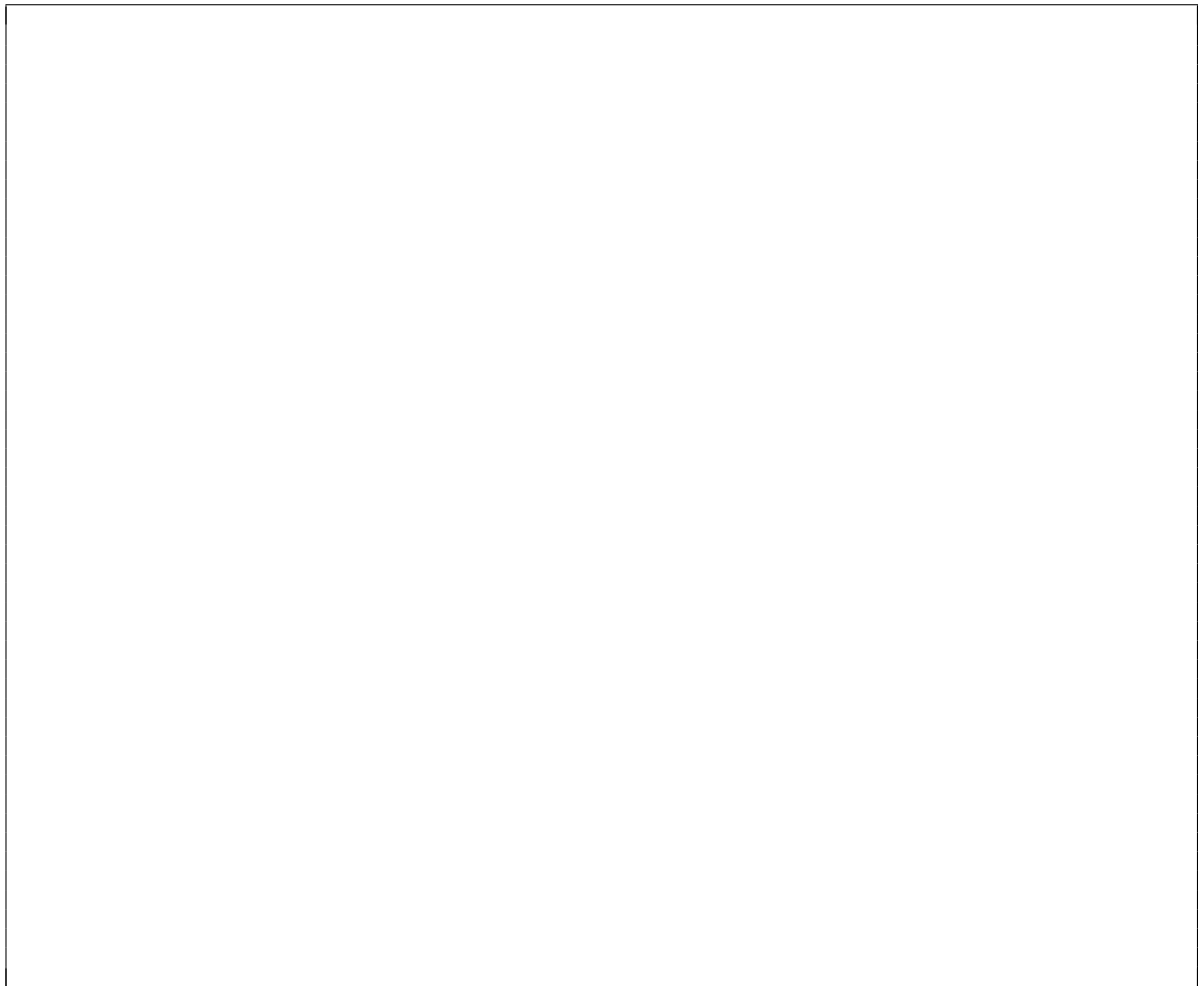
(3) The moment of inertia of the cylinder around the z -axis is:

6. (cf. Shifrin 7.4.21) Suppose that $R \subset \mathbb{R}^2$ is any region in the $x > 0$ halfplane and Ω is the corresponding region in $\mathbb{R}^3$ created by revolving R around the y -axis. One possible example for a very simple R is shown below.

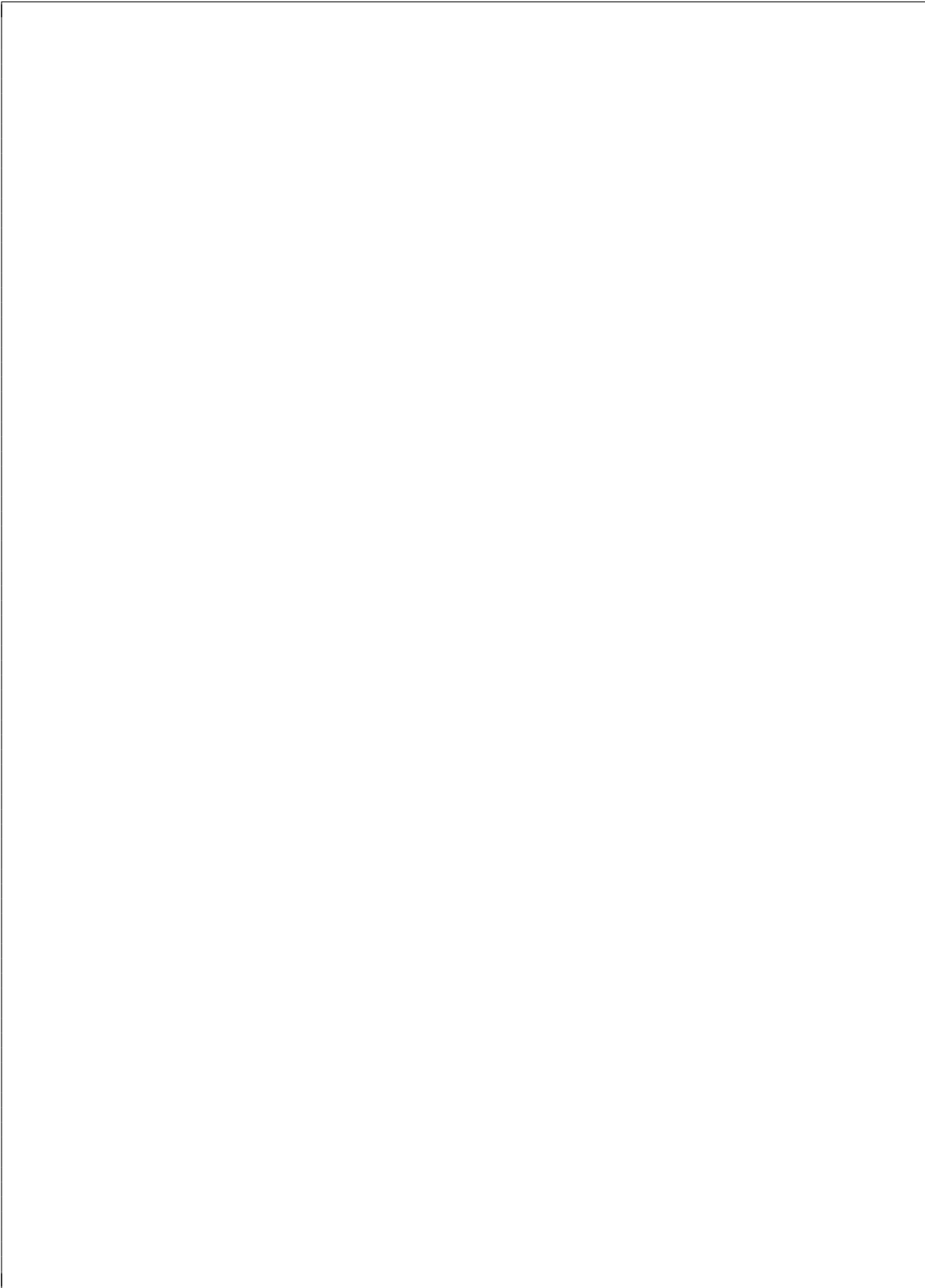


Prove *Pappus' Theorem*:^c if $\bar{x}$ is the x -coordinate of the center of mass of R , then

$$\text{Vol}(\Omega) = 2\pi \bar{x} \text{ area}(R).$$



^cThis theorem is more subtle than you might give it credit for, because the region R in the plane is arbitrary. To give a rigorous proof, you'll have to go back to the definition of integration over a region from Section 7.1



7. (cf. Shifrin 7.4.25) Prove or give a counterexample: The gravitational force on a test mass of any body $\Omega \subset \mathbb{R}^3$ with total mass M is the same as that of a point mass M located at the center of mass of Ω .

